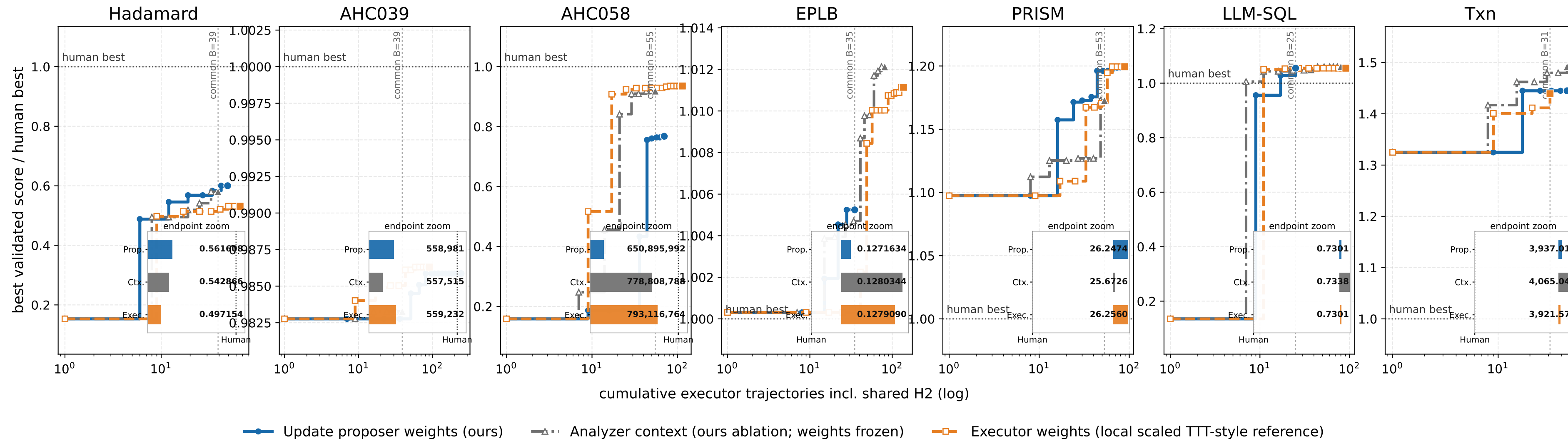


Where should test-time reward go? Proposer weights vs. analyzer context vs. executor weights



Reward route	Inference per feedback batch	Local weight update
Proposer weights (ours)	incumbent + bounded feedback -> target K=8 H1 proposals on one trained-phi replica -> frozen-executor trajectories (4 GPUs reserved)	terminal reward -> task-local harness ratchet + proposer LoRA r64/a128 (116.4M); 3 epochs; GBS 8; adapter continues
Analyzer context (ours ablation)	bounded feedback -> 2 frozen-model analysts after the cold round -> target K=8 frozen-proposer H1 proposals -> frozen-executor trajectories	terminal reward -> task-local harness/context ratchet only no gradient (proposer and executor weights frozen)
Executor weights (reference)	PUCT parent -> target K=8 fixed-harness executor trajectories (no H1 proposer or analyzer)	adaptive-beta LOO -> final-edit replay -> executor LoRA r32/a64 (58.2M); 1 epoch; GBS 4; adapter continues

Fairness: x = one shared fixed-H2 20-eval measurement plus actual launched executor trajectories; every fresh route starts from task.initial_program and gains count only after a batch finishes.
Readout: dashed = largest task-local budget shared by all routes. Y-axes are independently zoomed but retain score/human-best units; efficiency uses common-budget score/AUC, not endpoint pixel height.
Compute: recorded model/evaluator calls (lower bounds), sandbox time, optimizer boundaries, and sacct GPU-hours are separate ledgers. Failed/rejected/discarded operational work remains charged; proposal concurrency differs, so rollout efficiency is not FLOP efficiency.
Validation: endpoints are observed three-transition plateaus or explicit caps (never extrapolated, not absolute limits); black inset tick/whisker = program re-evaluation mean +/- 1 SD (N>=5). One unpaired campaign/route gives no campaign-level CI.
Scope: seven preselected strength tasks, not a population sample; blue inset is the locally ledged reported endpoint (unledged AHC039 559,534 excluded).
Reference: executor update is a local budget-scaled TTT-Discover-style baseline, not an official reproduction; final-edit replay, optimizer cadence, and K=8 differ from the published full-trajectory K=512 update. Hadamard/Txn reference qualifications are stated in text.